



CHAPTER

12

Kinetic Theory

Sr. No.	Concept	Formulas	Other information
1.	Ideal or perfect gas equation	Perfect gas equation; $PV = nRT$ Boltzmann's constant, $k_B = \frac{R}{N}$ $R = 8.31 \text{ J mole}^{-1} \text{ K}^{-1}$ $k_g = 1.38 \times 10^{-23} \text{ J K}^{-1}$	P = pressure of gas. V = volume of gas. R = universal gas constant N = Avogadro number n = number of moles T = temperature (in kelvin) of gas.
2.	Gas laws	Boyle's law: ($n, T = \text{constant}$) $PV = \text{constant}$ or $P_1V_1 = P_2V_2$ Charles law: ($n, P = \text{constant}$) $V \propto T$ or $\frac{V_2}{V_1} = \frac{T_2}{T_1}$ Gay Lussac's law: ($n, V = \text{constant}$) $P \propto T$ or $\frac{P_2}{P_1} = \frac{T_2}{T_1}$	P_1, P_2 = initial and final pressure respectively. V_1, V_2 = initial and final volume respectively T_1, T_2 = initial and final temperature respectively
3.	Pressure exerted by a gas	Pressure exerted by a gas, $P = \frac{1}{3}nmv_{rms}^2 = \frac{1}{3}\frac{M}{V}v_{rms}^2 = \frac{1}{3}\rho v_{rms}^2$ Here, $v_{rms} = \sqrt{\frac{3P}{\rho}}$	v_{rms} = root mean square velocity of gas. ρ = density of the gas P = pressure M = molecular mass m = mass of one molecule. n = number of molecules. V = volume of the gas.
4.	Avogadro's Law	At same temperature and pressure, equal volume of all gases contains equal number of molecules, i.e., $N_1 = N_2$	N_1, N_2 = number of molecules in the two gases.

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5.	Dalton's Law	The pressure exerted by mixture of non-reactive gases; $P = P_1 + P_2 + P_3 \dots$	P_1, P_2, P_3 = partial pressure exerted by individual gases.
6.	Average kinetic energy of a gas	Mean K.E. per molecule of a gas, $\bar{E} = \frac{1}{2} m \bar{v}^2 = \frac{3}{2} k_B T$ Mean K.E. per mole of a gas, $E = \frac{1}{2} M v_{rms}^2 = \frac{3}{2} RT = \frac{3}{2} k_B NT$ Relation between pressure and mean kinetic energy of gas; $PV = \frac{2E}{3}$	k_B = Boltzmann constant. R = universal gas constant N = Avogadro number m = mass of the gas
7.	Avogadro's number	Avogadro's number = $\frac{\text{Molecular mass}}{\text{Mass of 1 molecule}}$ or $N = \frac{M}{m}$	$N_A = 6.0225 \times 10^{23} \text{ mole}^{-1}$
8.	Average speed	Average speed, $\bar{v} = \frac{v_1 + v_2 + v_3 + \dots + v_n}{n}$ $\bar{v} = \sqrt{\frac{8k_B T}{\pi m}} = \sqrt{\frac{8RT}{\pi M}}$	$v_1, v_2, v_3 \dots v_n$ = velocities of the gas molecules n = number of molecules T = temperature of the gas M = molecular mass m = mass of molecule
9.	Root mean square speed	R.M.S. speed, $v_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 + \dots + v_n^2}{n}}$ $v_{rms} = \sqrt{\frac{3k_B T}{m}} = \sqrt{\frac{3RT}{M}}$	M = molecular mass
10.	Most probable speed	Most probable speed, $v_{mp} = \sqrt{\frac{2k_B T}{m}} = \sqrt{\frac{2RT}{M}}$	k_B = Boltzmann constant. T = temperature in kelvin
11.	Degrees of freedom	The number of degrees of freedom; $f = 3N - k$	N = number of particles in the system, k = number of independent relations between the particles

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12.	Law of equipartition of energy	Energy associated with each degree of freedom per molecule $= \frac{1}{2} k_B T$	T = temperature k_B = Boltzmann constant.
13.	Degrees of freedom, specific heats of monoatomic diatomic, and triatomic gases	Energy associated with 1 mole of gas, $U = \frac{f}{2} RT$ $C_V = \frac{f}{2} R$, $C_P = \left(1 + \frac{f}{2}\right) R$, $\gamma = \frac{C_P}{C_V} = 1 + \frac{2}{f}$ For monoatomic gas $f = 3$, so (3 Translational) $U = \frac{3}{2} RT$, $C_V = \frac{3}{2} R$, $C_P = \frac{5}{2} R$, $\gamma = 1.66$ For a diatomic gas, $f = 5$ (3 Translational + 2 Rotational) $U = \frac{5}{2} RT$, $C_V = \frac{5}{2} R$, $C_P = \frac{7}{2} R$, $\gamma = 1.4$ For a triatomic gas of non-linear molecules $f = 6$, so (3 Translational + 3 Rotational) $U = 3RT$, $C_V = 3R$, $C_P = 4R$, $\gamma = 1.33$	U = energy of one gram mole of gas f = degrees of freedom C_V = specific heat at constant volume C_P = specific heat at constant pressure R = universal gas constant T = temperature
14.	Mean free path	Distance of mean free path; $\bar{\lambda} = \frac{1}{\sqrt{2} n \pi d^2}$	n = number density d = diameter of the molecule.